Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI-assisted Thermal Design Tool for Electronic Manufacturing by Integration of Double-Radiation-Cooling Technology**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Mechanical Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	Design Tool	Metal Materials	Mechanical/ Electronic Component Materials Technology
Concept of Proposed Technology	Temperature is a primary cause of failure in electronic systems, and efficient heat dissipation in the vicinity of failure-prone chip packages is therefore essential for reliable operation. The project focuses on the technical development of innovative heat sink technology in conjunction with AI-assisted thermal design. Ultimately, both technical offerings will be commercialised in international electronics markets. Solutions are developed in the following <u>key areas:</u> 1. <u>Material and Technology for electronic heat sink</u> – The potential Korean partner company has developed an innovative technology for heat sink components for electronic systems. The new material and its specific surface technology have the potential to disruptively replace conventional Aluminium-based heat sinks. Metal-material technology results in doubling radiation (Radiation efficiency over 0.95). Heat radiation instead of conduction leads to disruptive potential of higher thermal performance with lower volume & decreasing weight. The double-radiation-cooling			

technology will be fully developed, adapted to electronics and finally evaluated.

2. Thermal Design tool for disruptive Technology by Modelling and Simulation - The introduction of new technologies in products requires convincing, visible technical advantages. We will create this in the first iteration with a model and simulation that incorporates this innovative technology. This will result in comparisons between conventional cooler designs and double-radiation-cooling. The result will be a comparison of performance parameters, and the advantages will be convincingly visible.
3. AI-Assistance for quick and efficient design for thermal management in electronics - Building on the initial thermal design model, an AI-assistance model will be created. This makes it possible to quickly calculate a wide range of electronic design variants and provide them in a comprehensible manner. Variants of the cooler architecture on real electronic assemblies can thus be very quickly evaluated virtually in terms of performance. This allows the final product designer for electronics to save development time and costs significantly. Furthermore, the acceptance of double-radiation-cooling is substantially increased.
4. Evaluation and validation on real electronics - The results of the thermal design modeler are evaluated by measurements on real electronic assemblies. For the model to be accepted, a validation step must be carried out, in which the actual physical thermal effects are compared with the calculated results of the simulation. Finally, the AI-assisted thermal design modeler will be proven and evaluated at electronics system level (at pilot customer)

Technological foundation

- Recent thermal management advances highlight rising interest in radiation-based cooling, with patents and research focused on dynamic radiative systems, engineered materials, and electronic integration. Companies and institutes are developing radiative cooling structures, vapor chambers, and advanced materials to enhance efficiency and reliability.
- the double-radiation-cooling technology of Korean SMEs, a next-generation heat-dissipation technology, enhances superior thermal radiation efficiency.
- double-radiation-cooling technology enables the development of highly advanced passive heat sink components, an advanced alternative to conventional heat sinks in semiconductor and electronic applications, offering distinct advantages over the traditional ones.
- New: AI-Assisted thermal design tool for electronic heat management based on innovative radiation technology enables acceptance in electronic designing companies

Main functions of the AI-assistance:

- The AI-Assisted thermal design tool considers both the heat source and its surrounding environment to deliver the most effective thermal dissipation strategy.
- Precise thermal analysis of the heat source characteristics (power density, operating temperature, material properties).
- Environmental adaptation, accounting for airflow, ambient temperature, enclosure constraints, and external conditions.
- Optimized radiation-based design, utilizing Advanced Passive Heat Sink technology for superior performance.

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	• AI-assisted modelling, ensuring reliable, efficient, and sustainable thermal solutions tailored to complex electronic systems. Potential uses: • Semiconductors and high-performance chips requiring efficient thermal management • Power electronic and mid-power systems (lighting, renewables, mobility) • Industrial and automotive electronics exposed to demanding environments • Next-generation AI and data centre hardware with high power density • Compact consumer electronics where space constraints limit convection cooling • Eco-friendly product designs emphasizing recyclability and reduced material dependency.
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* **Motivation**: Every technical unit contains and requires electronic systems, such for mobility, LED lightning, renewable energy supply etc. Electronic systems come with thermal losses due to computation by IC, switching electrical power or drive actors. Future chips and electronic systems increase computation power and much higher thermal losses must be managed. Cooling systems are required to avoid failures, electrical malfunction and degrading performance. Today heat sink systems, based on passive Aluminium component, are limited in performance, volume and weight. Innovative Korean Double-Radiation-Cooling is able to overcome these limitations. **Korean motivation**: Electronics, robotics, machinery manufacturing form the backbone of Korea's industrial landscape and represent key strengths in its export economy. It must be supported to keep the significance. Skilled workers and specialists are rare and AI-assisted tools for design and manufacturing of electronic products are required to fulfil future time-to-market demands and quality standards. The economical lead must be maintained through cooperation, advanced R&D and innovative materials, such as Double-Radiation-Cooling technology for thermal heat suppressors in electronics. Collaboration with Fraunhofer IKTS: The technology (Double-Radiation-Cooling) is already well known in the Asian economic area, and Korean companies are developing the first product solutions using it. However, detailed access to Europe, particularly Germany, has been limited. Fraunhofer's applied research profile and industrial network provides a unique gateway to the European market, enabling effective technology transfer, validation, and commercialization in Europe. **Need for joint R&D**: • Electronic devices failures occur due to insufficient thermal design, 55% of failures have origin in thermal loads. • Due to the advancement of semiconductor & electronic components, packages are becoming smaller but powerful. Heat-sink component for IC thermal management becoming bigger and heavier against high-density heating trouble. • Thermal heat management is mandatory for quality and reliability. • Thermal design efforts 15% costs in the value chain of electronics manufacturing. • Korean SME have developed an innovative <Advanced Passive Heat Sink> based on <double radiation heat technology> that has the potential to fundamentally change the thinking in the field of electronic heat management and design. • Coatings, functional layer and surface modification allows new heat sinks working by radiation instead by conduction maximized. • Technical advantages are multiple: higher heat efficiency (+120%), lower space requirement (-50%), lower total weight (-50%), lower system costs (-20%), optimal mechanical forming capability, reduce material effort (resources), reduce CO2 footprint Prototyping and laboratory testing have demonstrated the advantages and provided fundamental proof of concept. However, acceptance among potential users, i.e. electronics developers, is still lacking. To increase acceptance, Fraunhofer IKTS will develop an AI-assisted thermal design tool

	with the innovative features of Double-Radiation-Cooling technology. **Today's requirement**: • increase acceptance of disruptive heat sink technology (double-radiation) by modelling • Create thermal design support tool including double-radiation technology based properties • Create high performance AI-assisted thermal heat management tool for electronic devices for accelerated time to market and innovation included • Provide validated AI-assisted thermal design support tool with innovative heat sink technology for electronic product design.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. AI-assisted thermal design for electronics 2. automated optimization for competitive electronic design 3. disruptive heatsink material for power dissipation in electronics 4. Compact design flexibility for electronics 5. Sustainable and recyclable thermal material

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

Describe the specific technologies you aim to develop through this R&D project

The main objective of the project is to develop an AI-assisted thermal design tool for electronic systems that enables and accelerates the industrial adoption of disruptive double-radiation-cooling technology for advanced electronic cooling solutions.

To achieve the goal, the project team will carry out the following steps:

1. **Material-Design Coupling Research** (Korean SME)
   - Consolidate the technological maturity for the use of raw materials (aim TRL 8)
   - Incorporate the processing properties (pressing, bending, recyclability) of advanced passive heat sink into the design tool
   - Build a new library of radiation-based heat dissipation structures linked to material characteristics.
2. **Thermal Design Tool**
   - Analyze requirements for semiconductors, AI servers, automotive electronics and consumer devices
   - Create a modelling and simulation platform with respect to the disruptive double-radiation-cooling technology (TRL 5)
   - Preliminary adopting the customized thermal design modules tailored to pilot application
3. **AI-Assistance**
   - create a surrogate modelling-based thermal design tool by newly developed AI-assistance, with includes electronic thermal design, double-radiation-cooling technology and individual product parameters (TRL 7)
4. **Validation and Prototyping** (Korean SME)
   - Fabricate prototypes based on thermal design tool results and conduct experimental validation on real electronic components (TRL 6)
   - Benchmark performance against conventional heatsinks (temperature reduction, reliability, compactness, sustainability)
5. **System-Level Demonstration and External Validation**
   - Validate the solution at system level in an industrial environment with a pilot customer to build trust and enable commercialization (TRL 7).

The development of the design modeler with the specific advantages of the double-radiation-cooling technology enables consistent customer interaction. Customers recognize the potential of the new technology's performance through virtual comparison. For (SME) Korean companies, the German and European markets will be accessible through convincing technical arguments.

R&D Target Specifications

Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters

The target performance will be aimed at material-, modelling- and component-level, as follows:

Material - level / KPI

- HEAT radiation efficiency : over 0.95
- Reliable heterogenous junction : durable in automobile application
- Compatible for AEC-Q certification (Automotive Electronic Council – Qualification)
- Less than 50% of PCF comparing w/ Aluminum alloy plate

Modelling – level / KPI

The main target for the thermal design tool is to achieve an AI-assisted modeling/simulation for electronic/electrical thermal management in future electronic products. This will be iteratively realized in two main steps.

Step 1: Creation of thermal design platform with innovative, new, disruptive double-radiation-cooling

	Technology for heat suppression & dissipation. • The initial platform realizes the simulation tool for thermal design optimization in generic electronics packages with double-radiation-cooling technologies (first time). **KPI**: simulation based generic platform for variable heat source to cooling design is ready to use for pilot application **Step 2**: Creation of AI-assisted design support tool for thermal heat management in electronics, regarding the thermal parametric analysis in case of solid state as well as fluid state • Setup and training of AI-assistance as design support tool for thermal heat management in electronic product design • Design and implementation of modular AI architecture for thermal design support, including the selection of appropriate learning approaches (surrogate models, uncertainty estimation, and optimization), orchestration of multiple models for different design variants • Systematic training using synthetically generated thermal, geometric, and performance data from high-fidelity simulations • Validation of AI models through holdout tests, comparison against high-fidelity simulation data, and necessary experimental plausibility check **KPI**: validated AI-assistance for various electronic thermal designs is ready to use for product development (pilot customer finished successful) **Final Step:** Proof of AI-assisted design tool will be conducted through calculation of optimal thermal design by comparison of advanced heat sink based on double-radiation-cooling, in competition to conventional conduction-maximized heat sinks. Further, validation of optimized thermal design in electronics is done by measurements of thermal flux and temperature field analysis. **KPI**: increase of the trust and acceptance of advanced heat sink based on double-radiation-cooling technology and optimized electronic components. Component level / KPI • Higher thermal performance : 120% lower temperature of heat source • Smaller size/ volume : less than 50% • Light weight : less than 50% • Cost-effective forming : available profile by pressing & bending.
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* **Phase 1 – 01 to 06 months** **WP 1 – technical draft and conception of thermal design modelling tool (01-06 months) – *M1*** • Research the lack of general design tools (Modeling & simulation) • Define the outline of design tool • Define the Methodology of AI assistance **WP 2 – organization, partner integration and collaboration, business design (01-06 months)** • Bidirectional exchange of staff • Exchange of knowledge, Methodology Know-How • Trust building activities • Business model design (initial draft) **Phase 2 – 07 to 30 months** **WP1 - Material prototyping, material characterization and database initiation (07-12 months)** • Material Specialty functional metal for heat doubling radiation • Measurement of Parameter : related thermal influence • Formulation : DB/Library implementation. **WP 2 – initialization modelling and simulation platform for electronic (07-12 month) – *M2*** • Definition of generalized electronic design • Creation of simulation model with double radiation heat technology material properties • Testing and evaluation of the model based on thermal and heat flux measurements • Testing of thermal design tool on pilot electronics

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	WP 3 – AI-Assistant realization (10-24 month) – *M3* • Parametric Analysis : Solid state / Fluid state (AI assisted) • Structure Modeling : Optimal geometry & structure (AI assisted) • Model training and validation, Improvements • Thermal Simulation : component level & system level (AI assisted) **WP 4 – Evaluation and validation of AI-assisted thermal design tool (13-27 month)** • Sampling : press mold & welding • Evaluation : comparison with the simulation (Go back to analysis if need) • Validation : reliability & durability test **WP 5 – Verification at system level prior commercialization (25-36 month) – *M4*** • Filed verification : system level with pilot customer • Implementation for process : Digital twin (AI assisted) • Derivation : Electronics / Mechanics / active cooling **Milestones** ***M1*** – 06 month – Conceptualized design modeler, synchronized knowledge, trust building, business understanding ***M2*** – 12 month – Simulation platform is ready, and potential AI-assistance strategy is prepared ***M3*** – 24 month – AI-assisted design tool is developed, and internal verification plan is drawn ***M4*** – 36 month – Evaluation and validation of AI-assisted thermal design modeler ready, verified heat sink design results on electronic system at pilot application (customer)

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc* **Profile:** The Fraunhofer IKTS, with its department "Computer-Assisted Reliability and Material of Electronics" comes with very long-term experiences and personal expertise in the field of • Simulation based design for reliability of electronics • Thermo-mechanical material characterization of packaging materials in electronics • Performance and degradation measurement by Insitu-techniques in accelerated thermal and mechanical tests • Data driven prognostic and AI-assisted design for electronics Various pear reviewed publications substantiate and demonstrate the scientific experiences [1-12]. **Role:** IKTS creates a highly accurate & AI-assisted design tool for modeling & simulation: • Including parametric analysis in solid state and in fluid state • Modeling & simulation together • Related to production technology • Validated by measurements IKTS realizes design support to improve the performance of heat-sinks based on double-radiation-cooling in order to strengthen confidence in disruptive radiation founded cooling technologies. It supports the field verification of Korean's partner Materials and is ready to move commercialization by optimization for advanced demands with pilot customers. Within the project the proof of efficiency of high advanced heat sink based on double-radiation-cooling technology at system level (TRL 7-8) is targeted. Applicants of the design tool, such as electronic producers, designers and manufacturers, will be convinced of the material and design advantages. The simulation and the AI-design support tool will be positioned as key technology for proof, trust building and conviction of potential customers. **Door opener and multiplicator:** The applied science of IKTS activity leads to various industrial customers. The worldwide customer target market are power electronics and medium size power electronics for AI implemented system, robotics, industrial, renewable energy, automotive, and medicine application. Fraunhofer IKTS provides access to the German industrial and automotive electronics suppliers. Further the European network of Fraunhofer enables Korean companies to increase European market shares with HEAT-suppressor based on double-radiation-cooling products.
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.:* **Profile:** Korean SME which provides subcomponents for electronic products/ market and requires support in modelling and simulation by modern surrogate models with AI-techniques. **Role:** Targeting Commercialization & Standardization • Providing raw material and forming & sampling as a heat sink (as predicted by design tool) • Comparison with modeling & actual sample for the high accuracy • Related production technology • Supporting certification and standardization efforts to facilitate adoption in global semiconductor and electronics industries. **Need for international collaboration**: The collaboration between Koreans SME and Fraunhofer enables the market access and reduces the economic risk to enter European and German market. Merging competencies of Modelling/Simulation and double-radiation-cooling leads to validated AI-assisted Thermal Design Tool, which will be the door opener for European market. Cooperation between parties with leading technologies in specialized areas promotes the development of advanced solutions to technical challenges in the field of semiconductors/electronics.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Multiple effects and expected outcomes** will refer to Korean SME companies, technical R&D progress and climate and society effects. Marketing positioning • The project positions heat sink based on double-radiation-cooling technology as a disruptive alternative to conventional heat sinks, enabling a Korean SME to stand out in highly competitive global electronics markets. • By integrating validated simulation tools with AI-assisted design, the technology builds trust and credibility among producers, designers, and manufacturers, accelerating adoption. Economic Outcomes • According to Global Growth Insights, the global Aluminium heatsink market was valued at approximately USD 9.85 billion in 2024 and is projected to reach USD 14.25 billion by 2033. In comparison, the global copper heatsink market stood at USD 1.58 billion in 2024, with expectations to grow to USD 2.19 billion by 2033. This indicates that Aluminium heatsinks represent a market more than six times larger than copper. • In the existing heat sink market, the technology-owning companies will progressively expand the global market share - driven by the AI-assisted thermal design tool - to 30% by 2030. Target Markets • Power and medium-size electronics for AI systems, robotics, industrial automation, renewable energy, robotics, automotive, and mobility. • Consumer applications such as white goods, lighting devices, battery management systems, and LED displays. • Expansion into European and German markets via Fraunhofer's industrial network, enabling the Korean SME to increase market share and reduce entry risk. Technical benefits and outcome • Design support for increasing proof of high-performance HEAT-suppressor based on double-radiation-cooling to increase trust in disruptive heat suppressor technology • Field verification and ready to move commercialization by optimization for advanced demands with pilot customer who leading that application • Conviction of electronic producers, designers and manufactures • Proof of efficiency of HEAT-suppressor based on double-radiation-cooling Technology at product level (TRL 8) • Simulation and design support tool as key technology for proof, trust building and conviction of potential customers • Thermal management of power electronics must be controlled as a key factor and competency to create competitive electronic products The pace of innovation and product competition from Chinese and international companies in Korea's preferred industries is strong on international markets. AI-assistance product development will be one key factor to be competitive in future for electronic thermal design. Further cost reducing effects • Reduced time to market for electronic product development • Virtual product design planning with respect to physical functionality • Virtual testing reduces costs, resources and qualification time General climate and society effects Electrification by electronics devices is crucial to fulfil society's demands regarding climate change. E.g. transport & industry electrification with renewables and hydrogen. Korea, Germany and the EU are taking responsibility and following initiatives underlines it: • European regulation on CO2 reduction, waste reduction and resource savings require innovation Materials, as double-radiation-cooling which is functional metal and its applied

	product • Carbon Neutrality: Korea/Japan 2050, China 2060, India 2070. • European Green Deal (2019) - Legally commits the EU to climate neutrality by 2050. • Requires 50–55% emission reduction by 2030 and integrates circular economy, energy efficiency, and resource reduction as core strategies. • ETS & Carbon Tax: Korea runs ETS; Japan/China introduced schemes; ASEAN early stage. • EU Eco-design Regulation & Digital Product Passport - ensures products are designed for energy efficiency, durability, and repairability. • Proof by Digital Product Passport provides transparency on materials, recyclability, and carbon footprint. Is dominantly to be applied to electronics, textiles, batteries, and other resource - intensive products. • Circular Economy: Korea Green New Deal, Japan resource circulation, China recycling & renewables. • Product Standards: Korea eco-design, Japan energy efficiency, China green certification

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization